

AI Chatbot for Internal Helpdesk

(B.Tech CSE – AIML)



Introduction

- ❑ AI Chatbot for Internal Helpdesk is an intelligent web-based application.
- ❑ It automates responses to frequently asked employee questions.
- ❑ The chatbot uses Natural Language Processing (NLP) and Machine Learning.
- ❑ It helps reduce the workload of HR and IT support teams.
- ❑ The application is developed using Python and Flask.

Problem Statement & Objectives

❑ Problem Statement

Employees repeatedly ask similar questions.

Manual responses consume time and resources.

Delayed responses reduce productivity.

Organizations need an automated support solution.

❑ Objectives

Automate employee support.

Provide instant responses.

Implement NLP and Machine Learning.

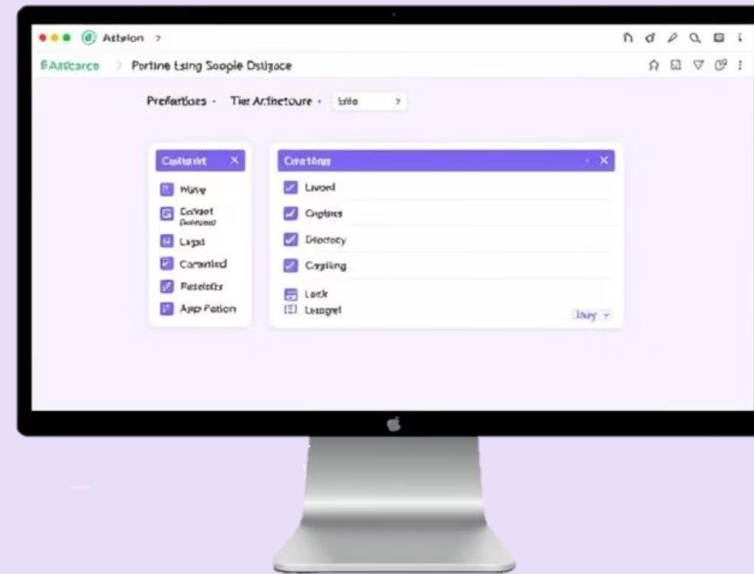
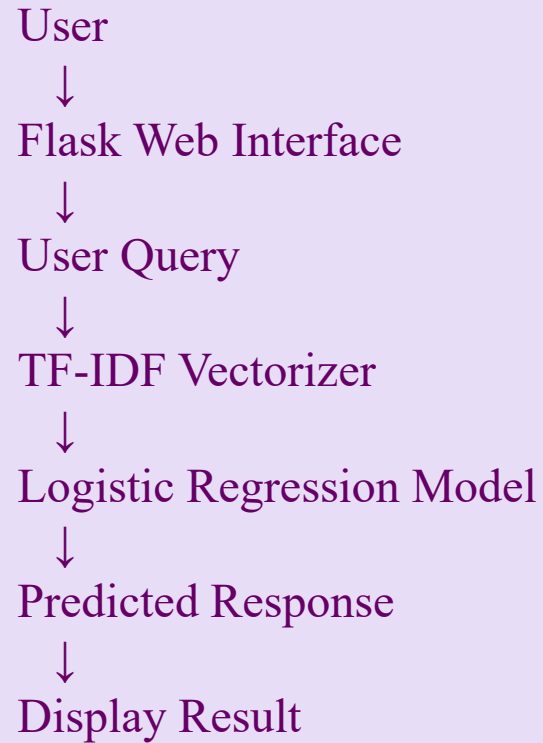
Develop a Flask-based chatbot.

Technology Stack

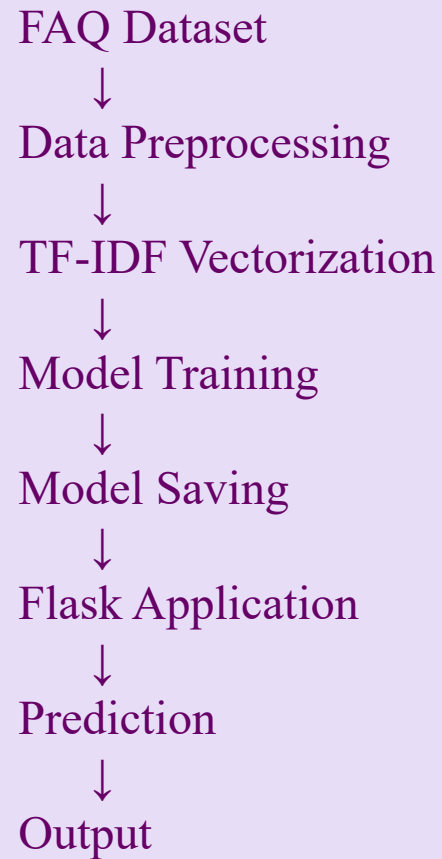
Technologies Used

- Python
- Flask
- Pandas
- Scikit-learn
- TF-IDF Vectorizer
- Logistic Regression
- Joblib
- HTML
- CSS
- Git & GitHub

System Architecture



Machine Learning Workflow

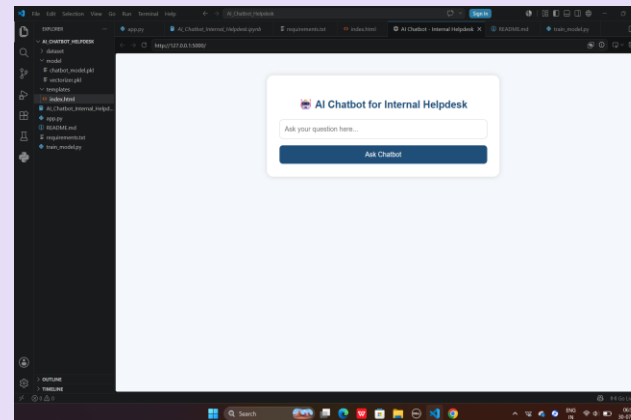


Project Implementation

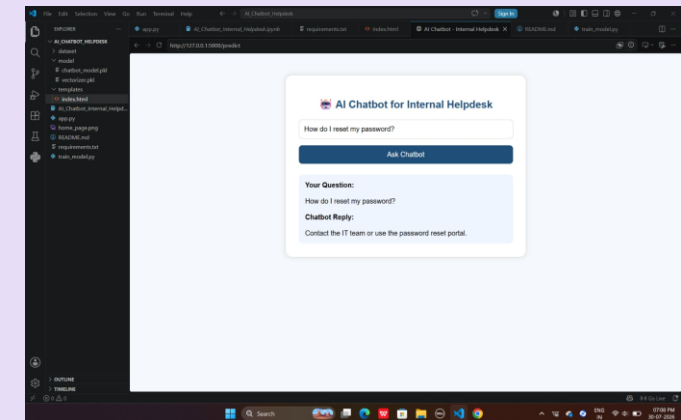
Project Modules

- ❑ FAQ Dataset
- ❑ Data Preprocessing
- ❑ TF-IDF Feature Extraction
- ❑ Logistic Regression Model
- ❑ Flask Web Application
- ❑ Prediction System
- ❑ User Interface

Home Page



Password Reset Response



Results & Advantages

Results

- ☐ Successfully predicts FAQ responses.
- ☐ Fast and accurate chatbot.
- ☐ User-friendly interface.
- ☐ Flask application works successfully.

Advantages

- ☐ Saves time
- ☐ Reduces manual work
- ☐ Easy to update
- ☐ Improves productivity
- ☐ Simple deployment

Future Scope

Future Enhancements

- ☐ Voice-enabled chatbot
- ☐ Multi-language support
- ☐ Database integration
- ☐ Admin dashboard
- ☐ Cloud deployment
- ☐ Authentication system
- ☐ Deep Learning models
- ☐ Integration with Microsoft Teams / Slack



Conclusion

- ❑ Successfully developed an AI-based Internal Helpdesk Chatbot.
- ❑ Automated responses to frequently asked employee queries.
- ❑ Applied NLP, Machine Learning, and Flask technologies.
- ❑ Improved response time and reduced manual support effort.
- ❑ Demonstrated a practical AI solution for workplace automation.

THANK YOU!